

Related work: implementations of quasi-local horizon spin and of the ingredients of Korzyński’s conformal method

Notes for the `KorzyńskiSpin.jl` package

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Abstract

We survey the literature for existing implementations of Korzyński’s conformal-decomposition definition of quasi-local horizon angular momentum [1], i.e. the pipeline: uniformization of the horizon two-metric (e.g. by Ricci flow), construction of the Möbius generators, and evaluation of the invariants $\vec{J}$ and $\vec{K}$. To the best of our knowledge *no published paper or public code implements this method end-to-end*; its citation record is small and consists of theory papers and of codes that adopted different spin definitions. What does exist falls into three groups: a methodologically adjacent (but different) spectral method by Jasiulek; the approximate-Killing-vector and Killing-transport spin measures implemented in the SpEC and Einstein Toolkit production codes; and general-purpose surface-uniformization software from computational conformal geometry. Consequently `KorzyńskiSpin.jl` would, as far as we can tell, be the first implementation, and validation must rely on analytic limits (Kerr, tilted slicings) and on cross-checks against the other spin definitions on the same data.

1 Summary

A search of the citation record of [1] (via Semantic Scholar and arXiv full-text search, June 2026) finds only a handful of citing works, none of which implements the method: they are reviews [13, 14], theoretical developments [10, 11, 12], and numerical-relativity papers that cite it as an alternative while implementing approximate-Killing-vector measures instead [5, 6]. Notably, the follow-up paper announced in [1] itself (“in preparation”, meant to contain the angular momentum flux law and the first law for non-axisymmetric horizons) appears never to have been published, and we found no numerical implementation by the author either.

The remainder of these notes describes the closest relatives, ordered by proximity to the present algorithm.

2 Closest relative in spirit: Jasiulek’s invariant-integral method

Jasiulek [2] computes quasi-local spin, mass and higher multipole moments on marginally trapped surfaces in 3 + 1 simulations with a pseudospectral, spherical-harmonic-based method that — like the present algorithm, and unlike Killing-vector approaches — avoids solving the Killing equation. The mechanism is however entirely different: spin is extracted from *surface moments of curvature invariants* (the scalar 2-curvature and the rotational Weyl scalar $\text{Im}\Psi_2$), e.g. central moments $\mu_n(\cdot) = \langle (\langle \cdot \rangle - \cdot)^n \rangle$, which are related algebraically to the isolated-horizon multipoles and inverted for the spin. No conformal decomposition, Ricci flow, or Laplacian-eigenfunction construction is used (an adapted-coordinates transformation is introduced only for axisymmetric checks), and [1]

is not cited. No public code was released. For the `KorzynskiSpin.jl` test suite, the published Kerr and perturbed-black-hole numbers of [2] are nevertheless a useful independent comparison point.

Jasiulek and Korzyński later developed a spectral *embedding flow* [3] that isometrically embeds horizon two-metrics into Euclidean $\mathbb{R}^3$ (Weyl’s embedding problem, via a Weingarten-type flow). This is methodologically adjacent — a geometric flow on horizon data, solved with spherical-harmonic spectral methods — but it solves a different problem (isometric embedding rather than uniformization) and computes no angular momentum.

3 What production numerical-relativity codes implement instead

All widely used codes evaluate the same surface integral $J_\phi = -\frac{1}{8\pi G} \oint \omega(\phi) \epsilon$ that underlies [1], but with a different prescription for the rotation vector ϕ :

- **SpEC: approximate Killing vectors.** The Cook–Whiting construction [4] (further developed by Owen) finds the vector field minimizing a Killing-equation residual by solving a generalized eigenvalue problem on the horizon. It is the standard spin diagnostic in the SXS collaboration’s SpEC code, used e.g. for nearly extremal horizons [6] and for the systematic study of *spin direction* measures by Owen et al. [5]. Both papers cite [1] as an alternative definition but do not implement it.
- **Einstein Toolkit: QuasiLocalMeasures.** The `QuasiLocalMeasures` thorn [8] computes the Dreyer–Krishnan–Schnetter–Shoemaker isolated-horizon spin [7] with Killing vectors obtained by Killing transport, or with coordinate rotational vectors.
- **Flat-space conformal Killing vectors.** Caudill, Cook, Grigsby and Pfeiffer [9] used the rotational Killing vectors of the flat conformal background in initial-data spin measurements; [1] notes explicitly that this is equivalent to the conformal-decomposition definition in many special cases, but not in general (no invariance discussion, no gauge fixing of ω).
- **Rácz’s axial vector construction.** A recent alternative [10] constructs axial vector fields from “centre-of-mass” unit-sphere reference systems; interestingly it also imposes $\ell = 1$ -type conditions on reference functions on the sphere, conceptually close to the eigenfunction triple χ_i of the present algorithm. It cites [1], is purely analytic, and has no accompanying software that we could find.

4 Building blocks that exist as software

The individual ingredients of the algorithm are all available, just never assembled for this purpose:

- **Uniformization.** Discrete surface Ricci flow and spherical conformal parameterization are mature topics in computational conformal geometry (Gu, Luo and collaborators [15]; various public MATLAB/Python spherical-conformal-map tools). These solve exactly the “round representative plus conformal factor” problem, but for triangle meshes rather than spectrally, and know nothing of ω , $\vec{J}$, $\vec{K}$.
- **Spectral horizon machinery.** The $\bar{\partial}$ -formalism [16] and fast spin-weighted spherical-harmonic transforms (e.g. the `FastSphericalHarmonics` Julia package) provide the pole-regular derivative and quadrature infrastructure; the pre-1.0 prototype of this package implemented the curvature computation of a horizon two-metric in this framework but did not reach the flow, eigenfunction, or invariant stages.

- **The rotation one-form integral.** The integrand $K_{ij}\phi^i s^j$ is implemented in every isolated-horizon spin diagnostic [7, 8]; only the choice of ϕ differs.

5 Consequences for validation

Since no reference implementation of [1] exists, the test strategy of the companion document `algorithm.tex` cannot include a code-to-code comparison of the method itself. The available external anchors are:

1. analytic limits: Kerr ($J = Ma$, axis along the symmetry axis, $\vec{K} = 0$ after gauge fixing), Schwarzschild and gauge-distorted Schwarzschild ($J = 0$);
2. the tilted-slicing Kerr test proposed in [1] itself, which exercises the Hodge gauge fixing;
3. cross-checks against *other* spin definitions on identical data: approximate-Killing-vector spin (SpEC-style [4]), Killing-transport spin (`QuasiLocalMeasures` [8]), and Jasiulek’s published values [2] — agreement is expected for nearly axisymmetric horizons, with quantifiable differences in strongly distorted regimes where the definitions genuinely differ;
4. the balanced round metric and horizon multipoles of Ashtekar, Khera, Kolanowski and Lewandowski [11], which use the same conformal decomposition.ourgoulhon, Le Tiec and Casals [17] give the balanced conformal factor and the multipoles of the Kerr horizon in closed form, an external anchor for the uniformization and eigenfunction stages that goes beyond $J = Ma$.

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